

Supplementary Material: Reliability-Aware Adaptive Contention Control for EMLSR Multi-Link Operation in IEEE 802.11bn WLANs

Jiha Kim, Seunghyun Park, and Hyunhee Park, Senior Member, IEEE

This document is the supplemental material for the manuscript “Reliability-Aware Adaptive Contention Control for EMLSR Multi-Link Operation in IEEE 802.11bn WLANs.” Equation numbers of the form (5) refer to the equations in the main manuscript.

Appendix A Analytical Model Overview

Appendix B Supplementary Derivations

STR steady state. Under saturated STR, the tagged state $X_\ell(t) = (i, k)$ has countdown $\Pr\{(i, k-1) \mid (i, k)\}=1$, success-reset $\Pr\{(0, k) \mid (i, 0)\}=(1-p_\ell)/W_0$, and collision-advance $\Pr\{(i+1, k) \mid (i, 0)\}=p_\ell/W_{i+1}$ ($\Pr\{(m, k) \mid (m, 0)\}=p_\ell/W_m$ at the maximum stage), giving $b_{i,0}^{(\ell)}=p_i^\ell b_{0,0}^{(\ell)}$ ($i < m$), $b_{m,0}^{(\ell)}=p_m^\ell b_{0,0}^{(\ell)}/(1-p_\ell)$, and $b_{i,k}^{(\ell)}=(W_i-k)b_{i,0}^{(\ell)}/W_i$; normalising and summing $\tau_\ell = \sum_i b_{i,0}^{(\ell)}$ recovers (5), with (6) monotone in n_ℓ . EMLSR reduction. An exact description requires $\mathbf{Z}(t) = (X_{2.4}, X_5, X_6, A(t), R(t), U_{2.4}, U_5, U_6)$; we collapse this state space via the availability $\alpha_\ell = 1 - \beta_\ell$ with $\beta_\ell \approx \sum_j g_j H_{j \rightarrow \ell}$ ((26): a same-link success contributes only T_{tr} while an other-link success induces $T_{\text{tr}} + T_{\text{sync}}$), yielding the per-class success probabilities (30)–(33) as a validity-range approximation in $(T_{\text{tr}}, T_{\text{sync}}, n_\ell)$. Unsaturated extension. Eqs. (39)–(41) fix arrival/service/utilisation; under (42), throughput is $\min(\lambda, \mu)$ with (43) as the queue-delay closure and (44) as its M/M/1/K saturation bound. We therefore center quantitative claims on throughput, busy ratio, HOL-service, and saturation onset rather than total queue delay.

Appendix C Additional Robustness Sweeps

A. Asymmetric Multi-Band Rates

Section VII-B of the main paper showed that a symmetric MCS 7/20 MHz/1 SS configuration erases the MLO throughput advantage under EMLSR. To find when the link-diversity gain materialises, we swept per-link MCS triplets on 2.4/5/6 GHz at fixed 20 MHz/1 SS: symmetric (7, 7, 7), mild (5, 7, 9), and strong (3, 7, 11).

Three messages emerge from Fig. 2. First, mild asymmetry recovers the MLO gain: (5, 7, 9) matches or exceeds (7, 7, 7) throughout (STA 50 sim 95.6 vs. 93.0 Mbps, +2.8%) because the fast 6 GHz link clears its queue faster, reducing the cross-link blocking load $\sum_j g_j H_{j \rightarrow \ell}$ on the slower links. Second, strong asymmetry bottlenecks: (3, 7, 11) falls below the symmetric baseline at every STA (STA 50 sim 90.2 Mbps) as the slow 2.4 GHz link inflates $T_{s,2.4} + K \cdot T_{\text{MPDU},2.4}$ and $H_{2.4 \rightarrow \ell}$, blocking the other links during its long TXOP. Third, closure accuracy is scenario-agnostic (mean ana/sim gap for STA ≥ 5 : 4.32, 3.88, 3.18% for sym, mild, strong). The takeaway: modest per-band MCS diversity unlocks a small deterministic gain, while aggressive spread harms throughput unless balanced by proportional bandwidth or spatial-stream asymmetry.

B. Packet-Size Sensitivity

At the same saturated operating point we swept payload $\in \{500, 750, 1000, 1250, 1500, 1750, 2000\}$ B at STA $\in \{12, 24, 26\}$, with per-MPDU airtime $T_{\text{MPDU}} = L/R_{\text{phy}}$ from the MCS 7/20 MHz/1 SS PHY rate.

Fig. 3 shows: (i) throughput grows with payload ($\sim 58 \rightarrow 127$ Mbps at STA 12; $\sim 55 \rightarrow 117$ Mbps at STA 24/26) as fixed MAC/PHY overhead amortises; (ii) HOL-service delay rises 1.6 $\rightarrow$ 3.0 ms (STA 12) and 3.3 $\rightarrow$ 7.1 ms (STA 24/26) as larger MPDUs enlarge T_s ; (iii) busy ratio climbs $\sim 0.78 \rightarrow 0.90$. STA 24 and STA 26 curves nearly overlap, STA 12 sits above on thr but below on HOL/busy, confirming the closure separates payload L and contender n correctly; ana-sim agreement is $\sim 2\text{--}3\%$ on all three.

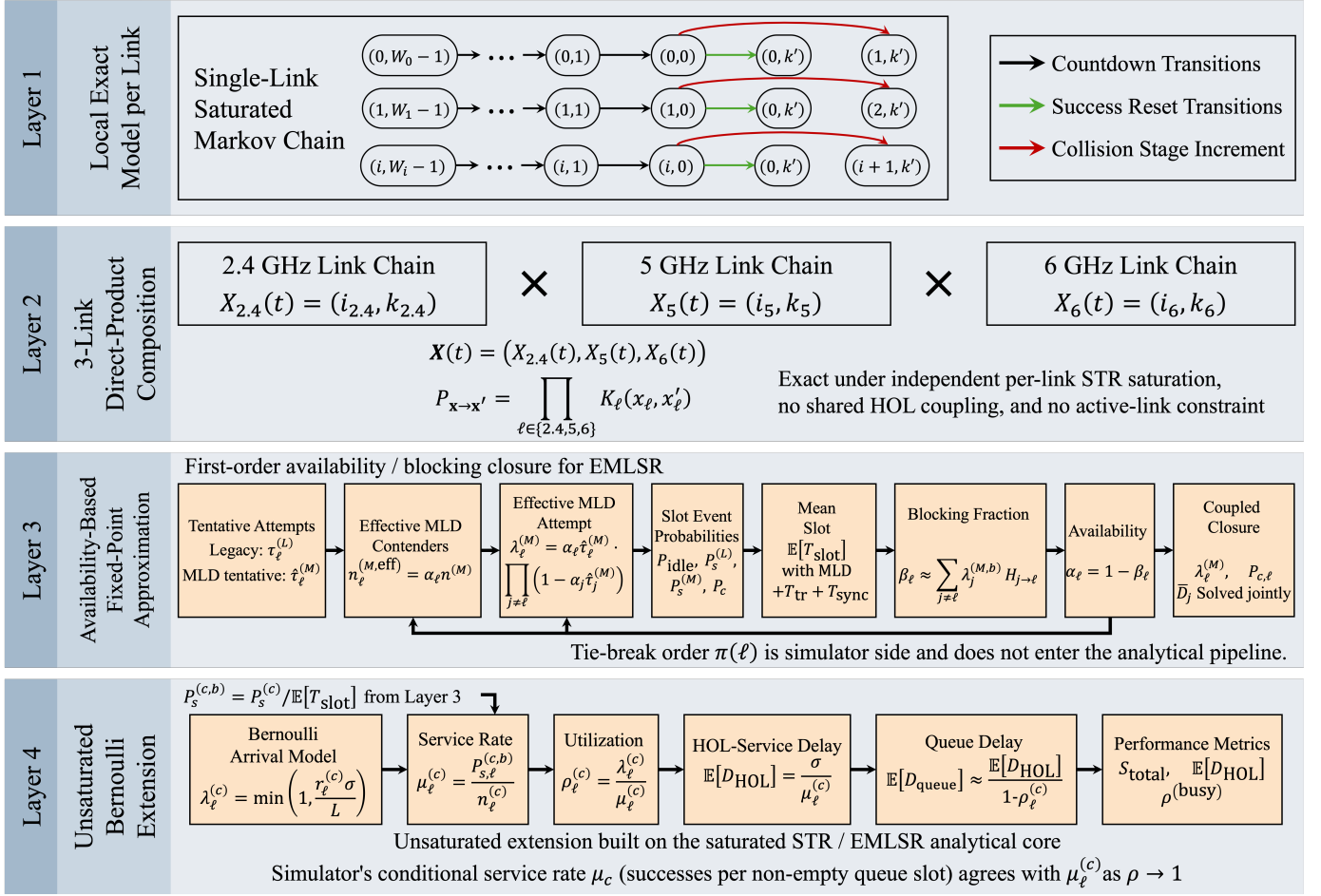


Fig. 1. Summary diagram linking exact STR, approximate EMLSR, and the unsaturated extension (four-layer analytical pipeline of the main paper).

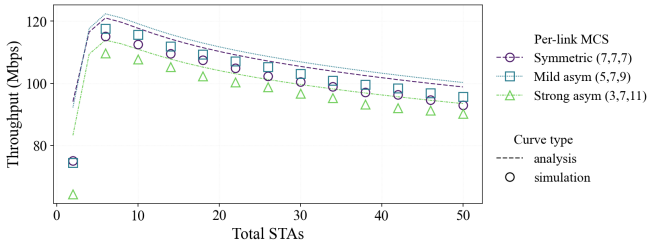


Fig. 2. Saturation throughput under per-link MCS asymmetry: sym (7, 7, 7), mild (5, 7, 9), strong (3, 7, 11).

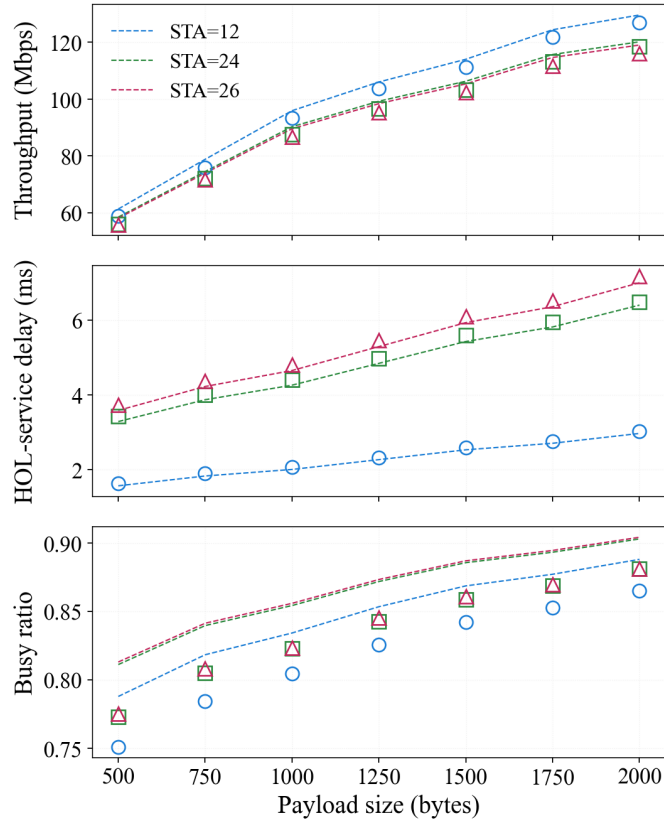


Fig. 3. Payload-size sensitivity at STA counts {12, 24, 26}, MPDU payload 500–2000 B.